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Assignment 1 Design Document

1. Rust collection data structures are polymorphic, so you will have to explain what each type variable (those are the T and V and S and K you see in the documentation) will actually be. The answer here should be a concrete type, such as `u64` or `String`
    a. The data structure is a HashMap, where keys are the fingerprints (strings), and the values are the fingerprint groups (Vector of strings).
    b. HashMap<String (fingerprint),Vec<String>>(fingerprint groups)
2. Please **describe the invariant** that will hold when you are partway through reading input lines.
    a. For every fingerprint with more than one occurrence, it exists as a fingerprint group in the hashmap.
    b. The HashMap contains unique fingerprints as keys
    c. The HashMap contains Vectors that are associated with each unique fingerprint, and contain names that share the same unique fingerprint.
    d. When input lines are processed, the names are put into the Vectors
3. Based on these two explanations, explain briefly how you will compute the fingerprint groups once all input has been read.
    a. After all the input lines have been read and processed, iterate through the HashMap, and filter out all fingerprints with only one associated name.

b. The hashmap now contains every fingerprint associated with vectors that contain more than one name, implying fingerprint groups are formed containing names with their associated fingerprint.
c. Using this, we can then print out the names of each fingerprint, sorted by their associated fingerprint group.